Homework 6

Olivia, Andre, Emmanuel

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
Warning: package 'dplyr' was built under R version 4.4.3
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.2.0     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   4.0.2     ✔ tibble    3.2.1
✔ lubridate 1.9.3     ✔ tidyr     1.3.1
✔ purrr     1.0.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
Warning: package 'knitr' was built under R version 4.4.3
```

```
here() starts at /Users/andreleon/Desktop/edsthmwk/hw_6
```

```
Rows: 5748 Columns: 12
──                            Column                            specification
──────────────────────────────────────────────────────────────────────────────
Delimiter: ","
chr  (2): sex, frl
dbl (10): schidkn, reg_size, reg_size_aid, small_size, white, black, other, ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

We used the tidyverse (H. Wickham *et al.* [1]) , knitr (Y. Xie [2]) , and here (K. Müller [3]) packages as well as some additional information on the star data (E. Hanushek [4]).

References

```
 [1] "schidkn"      "sex"          "frl"          "reg_size"     "reg_size_aid"
 [6] "small_size"   "white"        "black"        "other"        "totexp"
[11] "tmathss"      "treadss"
```

```
`summarise()` has regrouped the output.
ℹ Summaries were computed grouped by sex and frl.
ℹ Output is grouped by sex.
ℹ Use `summarise(.groups = "drop_last")` to silence this message.
ℹ Use `summarise(.by = c(sex, frl))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Sex	FRL	Math Mean	Math SD	Reading Mean	Reading SD
boy	no	492.85	46.34	441.46	32.32
boy	yes	469.87	46.09	425.38	26.63
girl	no	501.21	45.96	448.54	34.52
girl	yes	477.51	46.30	430.80	27.42

#The summary statistics table above shows that students who were not eligible for free or reduced lunch had higher average math and reading scores than students who were eligible for free or reduced lunch. Girls had higher average scores than boys in both subjects across free and reduced lunch categories. The standard deviations shows variability in both math and reading scores.

Bibliography

[1] H. Wickham *et al.*, "Welcome to the tidyverse," *J. Open Source Softw.*, vol. 4, no. 43, p. 1686–1687, Nov. 2019, doi: 10.21105/joss.01686.

[2] Y. Xie, "knitr: A General-Purpose Package for Dynamic Report Generation in R." 2026. [Online]. Available: https://yihui.org/knitr/

[3] K. Müller, "here: A Simpler Way to Find Your Files." 2025. [Online]. Available: https://here.r-lib.org/

[4] E. Hanushek, "Some Findings From an Independent Investigation of the Tennessee STAR Experiment and From Other Investigations of Class Size Effects," vol. 21, 1999, doi: 10.3102/01623737021002143.